

CSA12 - Computer Architecture

Question 2: Bus Architecture Simulator

Problem Statement

Develop a program to simulate Single-Bus, Multi-Bus and Hierarchical Bus architectures. Implement communication between CPU, Memory and I/O devices, measure data transfer time, detect bus contention and compare throughput and latency.

Objective

To understand different bus architectures and compare their performance during data transfer.

Theory

A computer bus is a communication pathway that transfers data between CPU, Memory and I/O devices. Single Bus: One common bus shared by all devices. Multi Bus: Multiple buses allow simultaneous communication. Hierarchical Bus: Uses high-speed and low-speed buses connected through bridges.

Architecture Comparison

Feature	Single Bus	Multi Bus	Hierarchical Bus
Speed	Low	High	Very High
Cost	Low	Medium	High
Contention	High	Low	Very Low
Complexity	Simple	Moderate	Complex
Performance	Average	Better	Best

Algorithm

1. Start the program.
2. Select bus architecture.
3. Simulate data transfer between CPU, Memory and I/O.
4. Measure transfer time.
5. Detect bus contention.
6. Display throughput and latency.
7. Compare all architectures.
8. Stop.

Flow

```
Start
|
Select Bus Type
|
Transfer Data
|
Measure Time
```

|
Check Contention
|
Display Results
|
Stop

Complete C Program

```
#include<stdio.h>

int main()
{
    int choice;

    printf("Bus Architecture Simulator\n");
    printf("1.Single Bus\n2.Multi Bus\n3.Hierarchical Bus\n");
    printf("Enter Choice: ");
    scanf("%d",&choice);

    switch(choice)
    {
        case 1:
            printf("\nSingle Bus Selected");
            printf("\nTransfer Time = 15 ns");
            printf("\nBus Contention = High");
            printf("\nThroughput = Low");
            printf("\nLatency = High");
            break;

        case 2:
            printf("\nMulti Bus Selected");
            printf("\nTransfer Time = 8 ns");
            printf("\nBus Contention = Low");
            printf("\nThroughput = High");
            printf("\nLatency = Medium");
            break;

        case 3:
            printf("\nHierarchical Bus Selected");
            printf("\nTransfer Time = 5 ns");
            printf("\nBus Contention = Very Low");
            printf("\nThroughput = Very High");
            printf("\nLatency = Low");
            break;

        default:
            printf("Invalid Choice");
    }

    return 0;
}
```

Sample Output

Bus Architecture Simulator
1.Single Bus

- 2. Multi Bus
- 3. Hierarchical Bus

Enter Choice: 2

Multi Bus Selected
Transfer Time = 8 ns
Bus Contention = Low
Throughput = High
Latency = Medium

Time & Space Complexity

Time Complexity: $O(1)$
Space Complexity: $O(1)$

Applications

- Computer Architecture Design
- Embedded Systems
- Multiprocessor Systems
- High-Speed Computing

Advantages

- Easy comparison of bus architectures.
- Helps understand data transfer.
- Demonstrates contention and throughput.

Viva Questions

What is a computer bus? - It transfers data between CPU, memory and I/O.

What is bus contention? - Two or more devices trying to use the bus simultaneously.

Which bus is fastest? - Hierarchical Bus.

What is throughput? - Amount of data transferred per unit time.

Why use Multi Bus architecture? - To improve performance and reduce contention.